

Computer Security and Networks

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Outline of This Lecture

- ▶ Module Arrangements: When, Where, Who.
- ▶ Module Outline
- ▶ Module Outcome

Module Arrangements at Edgbaston

- ▶ Who: Eike, Pascal, and Rishi.

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- ▶ Where: On Campus lectures.
 - ▶ Monday, 15:00–17:00, Teaching and Learning Building LG18.
 - ▶ One hour on revisiting lectures, one hour pop quiz.

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- ▶ Microsoft Teams channel.

Module Evaluations

- ▶ 20% continuous assessment, 80% exams.
- ▶ Token based exercises on VMs.
 - ▶ When you complete an exercise on the VM you will usually find a token (or flag).
 - ▶ You submit the token to a website, to show you have solved the exercise.
 - ▶ Tokens are unique to your VM. You must not share VMs or Tokens.
- ▶ Make sure your computer has sufficient space; should be possible to run it from external disks.

DO NOT TRY OUT ANYTHING ON COMPUTERS YOU DO NOT OWN

- ▶ It is illegal to access computers without the owner's permission.
- ▶ Most access are logged, and it is easy to get caught.
- ▶ Trying things “just for fun” could be punishable offense.

Learning Outcome

- ▶ Understand basic concepts of cryptography and SQL
- ▶ Understand basic concepts of cloud services, in particular storage
- ▶ Demonstrate an understanding of the threats to data stored on a computer, locally or in the cloud
- ▶ Demonstrate an understanding of the threats to data sent on the network
- ▶ Identify risks and use techniques to eliminate or mitigate them.

Module Outline

- ▶ Cryptography
- ▶ Access Control
- ▶ Introduction to Networking
- ▶ Security Protocols
- ▶ Web Systems and Attacks
- ▶ Other Common Attacks and Defenses

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 - ▶ Control Systems.
 - ▶ Hardware devices.
- ▶ How do you safeguard: security goal, estimate impact of attacks, and design mitigations
- ▶ Analyse systems, spot vulnerabilities, build protection.

Information Security:Aims

- ▶ **Confidentiality:** Attacker should not retrieve any information.
- ▶ **Integrity and Authenticity:** Received data is authentic and the sender is genuine.
- ▶ **Availability:** Data should accessible on demand.

Information Security: Potential Attackers

Anyone and Everyone

Information Security: Potential Attackers

- ▶ **Hackers:** Potentially learning by running known attacks, exploiting vulnerabilities.
- ▶ **Criminals:** Take control of computers via bugs in softwares. Phishing attacks, Denial of Service (DoS attacks)
- ▶ **Governments:** Extreme computing powers, control on resources (wiretaps),...
- ▶ **Business Houses like ISPs:** Spying to sell your data.

Some Known Attacks

▶ **Ransomware:**

- ▶ Malwares: Trojan disguised as legitimate files.
- ▶ *Wannacry 2017* moved automatically via unpatched vulnerabilities in Microsoft Windows (Eternal Blue of NSA).
- ▶ The malware encrypted the data on the computer and asked for payments in bitcoins.
- ▶ Widespread impact, NHS and NISSAN among affected.

▶ **Phishing:**

- ▶ emails pretending to be from known people.
- ▶ emails asks for username and password and asks for software installation, includes word macros.
- ▶ install malware to spread within networks, downloads further malware.

Course Outcome: Informal

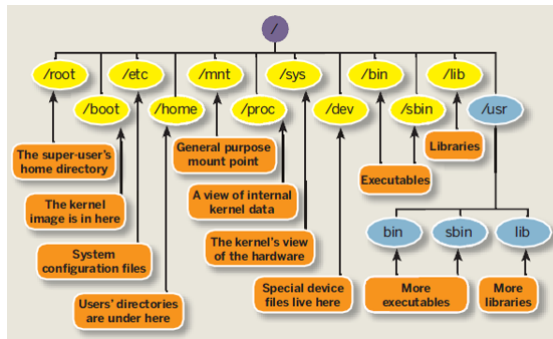
First steps on how to stay safe in the digital world.

Part II: Unix Commands and Shell Scripts

Why are we talking about these?

- ▶ shell commands: operations on files.
- ▶ utilising OS commands: potentially faster.
- ▶ **Batch Processing**: Sequentially running programs.

linux filesystem



Shell Commands

- ▶ structure: *command-name* *options* *input*—*output*
- ▶ echo, mkdir, ls, pwd, cat, cp, mv, rmdir, rm, touch, locate, find, grep, diff, du, head, diff, tar, top , history